

059 Safety and Efficacy of Donidalorsen in Adolescent Patients With Hereditary Angioedema: 1-Year Results From OASISplus



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RATIONALE: In the phase 3 OASIS-HAE study (NCT05139810), donidalorsen reduced hereditary angioedema (HAE) attack rates overall and in adolescent patients. Here, we report safety and efficacy results up to 1 year of the OASISplus study (NCT05392114) for adolescent patients who opted to switch from a prior HAE long-term prophylactic medication (LTP) to donidalorsen.

METHODS: Patients on stable doses (≥ 12 weeks) of berotralstat, intravenous (IV) or subcutaneous (SC) C1INH, or lanadelumab switched to donidalorsen 80 mg SC once every 4 weeks without washout per a prespecified algorithm. The primary endpoint was the incidence and severity of treatment-emergent adverse events (TEAEs). Other endpoints included HAE attack rate/month, Angioedema Quality of Life (AE-QoL), and plasma prekallikrein (PKK).

RESULTS: Six adolescent patients (age, 12–17 years) switched from a prior prophylactic HAE therapy to donidalorsen; 4 completed 1 year of treatment (C1INH: SC, n=2; IV, n=1; lanadelumab, n=1). Three of four (75%) patients reported TEAEs, none of which were severe, serious, related to donidalorsen, or led to treatment discontinuation. Through Week 52, mean HAE attack rates decreased by 85% from baseline (on their prior HAE LTP). Plasma PKK concentrations decreased by 66% over the same period. AE-QoL total scores improved by a mean of 4.8 points from baseline. Findings from 7 additional adolescent patients enrolled in a separate open-label extension cohort of OASISplus were consistent with those reported here.

CONCLUSIONS: In adolescents, donidalorsen had an acceptable safety and tolerability profile and improved HAE attack rates and quality of life, consistent with overall results from OASISplus.

060 Navenibart Demonstrates Durable Efficacy and Tolerability Across Biological Sexes: Subgroup Analysis from the ALPHA-STAR Trial



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RATIONALE: Hereditary angioedema (HAE) is a rare autosomal dominant disease characterized by recurrent swelling attacks. Biological sex influences symptom burden and disease severity. Here, we present data

evaluating sex-based response to navenibart, a long-acting monoclonal antibody plasma kallikrein inhibitor.

METHODS: ALPHA-STAR (NCT05695248), a Phase 1b/2 multicenter, dose-ranging, proof-of-concept trial, assessed the safety and efficacy of navenibart in adults with HAE. Participants received subcutaneous navenibart (3 cohorts): 450 mg (day 1), 600mg(day 1)/ 300mg(day 84), or 600mg/600mg (days 1 and 28) for a treatment period that lasted up to 9 months (6 months after the final injection). This post-hoc analysis evaluates response to navenibart by biological sex based on frequency and severity of attacks.

RESULTS: Of the 29 participants enrolled, 16 participants (55.2%) were female with a mean(SD) age of 45.8 (15.9), and 13 (44.8%) were male with a mean(SD) age of 47.2(18.1). Overall, 25(86.2%) were HAE Type 1. Treatment-emergent adverse events (TEAEs) reported by 25/29(86.2%) participants were distributed similarly (87.5% in females vs. 84.6% in males); related TEAEs occurred in 4 participants; all TEAEs were mild to moderate in severity. Females reported higher mean(SD) monthly baseline attack rates than males (2.60(1.77) vs. 1.77(0.81), respectively). Mean monthly attack rates decreased by 84.1% in females and 89.8% in males by the end of the treatment period.

CONCLUSIONS: Navenibart demonstrated consistent and durable efficacy with a favorable safety profile regardless of biological sex. These findings support the continued evaluation of navenibart (every 3- and 6-month) in a Phase 3 trial (ALPHA-ORBIT, NCT068428).

061 Long-Term Durability and Safety of Lonvoguran Ziclumeran (Lonvo-z, NTLA-2002) 50mg in Patients With Hereditary Angioedema



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RATIONALE: Lonvoguran ziclumeran (lonvo-z) is an investigational, CRISPR-based, one-time treatment designed to rebalance the kallikrein-kinin system in patients with hereditary angioedema (HAE) by permanently editing the *KLKB1* gene. Lonvo-z was evaluated in a dose-finding Phase 1/2 study (NCT05120830). After lonvo-z 50mg was identified as the Phase 3 dose, patients who received placebo or lonvo-z 25mg were eligible to receive lonvo-z 50mg. Herein, we describe longer-term durability/safety among all patients who received lonvo-z 50mg.

METHODS: Patients (N=28) received a single infusion of lonvo-z 50mg (initial dose 50mg, n=15; lonvo-z 25mg to 50mg [25-50mg], n=7; placebo to lonvo-z 50mg [PBO-50mg], n=6). Safety and efficacy outcomes were assessed.

RESULTS: As of 12Feb2025, median (range) follow-up after receiving lonvo-z 50mg was 18.9 (10.2-24.4), 5.6 (3.8-7.4), and 2.8 (0.5-3.6) months for 50mg, PBO-50mg, and 25-50mg, respectively. Following lonvo-z 50mg infusion, the mean (SD) monthly attack rate reduction from baseline to the latest assessment was 95% (13.2) without long-term prophylaxis (LTP). At latest follow-up, 93% (26/28) of patients were attack-free and LTP-free for a median (range) of 10.2 (0.5-24.0) months. The most common adverse events ($\geq 20\%$) were infusion-related reaction (50%), fatigue (32%), headache (21%), and upper respiratory tract infection (21%), all Grade 1/2 and generally transient. The safety profile for PBO-50mg and 25-50mg after lonvo-z 50mg was consistent with initial 50mg dose cohort. Updated data will be presented.

CONCLUSIONS: With up to 2 years of follow-up, a one-time treatment with lonvo-z 50mg led to robust, sustained reductions in HAE attacks without the need for LTP with an acceptable safety profile.